

TrES-3b

R-0590

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230818 · created 2026-07-24T23:07:25+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460174.7763 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.057
Transit depth (Rp/Rs)^2	3.59 +/- 2.18 [%]
Orbital Inclination (inc)	86.56 +/- 2.84
Airmass coefficient 1 (a1)	1.1 +/- 0.67
Airmass coefficient 2 (a2)	-0.064 +/- 0.065
Scatter in the residuals of the lightcurve fit is	56.0 %
Best Comparison Star	#2 - [499, 161]
Optimal Aperture	0.99
Optimal Annulus	5.54
Transit Duration (day)	0.0127 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.19 $\pm$ 0.057 vs 0.16309 (deviation 0.47 σ)
Duration fit vs published	0.0127 d vs 0.0567 d (deviation 8.3 σ)
Mid-time O-C	0.7 min
Depth SNR	3.3
Residual scatter	56.0 %

Light curve

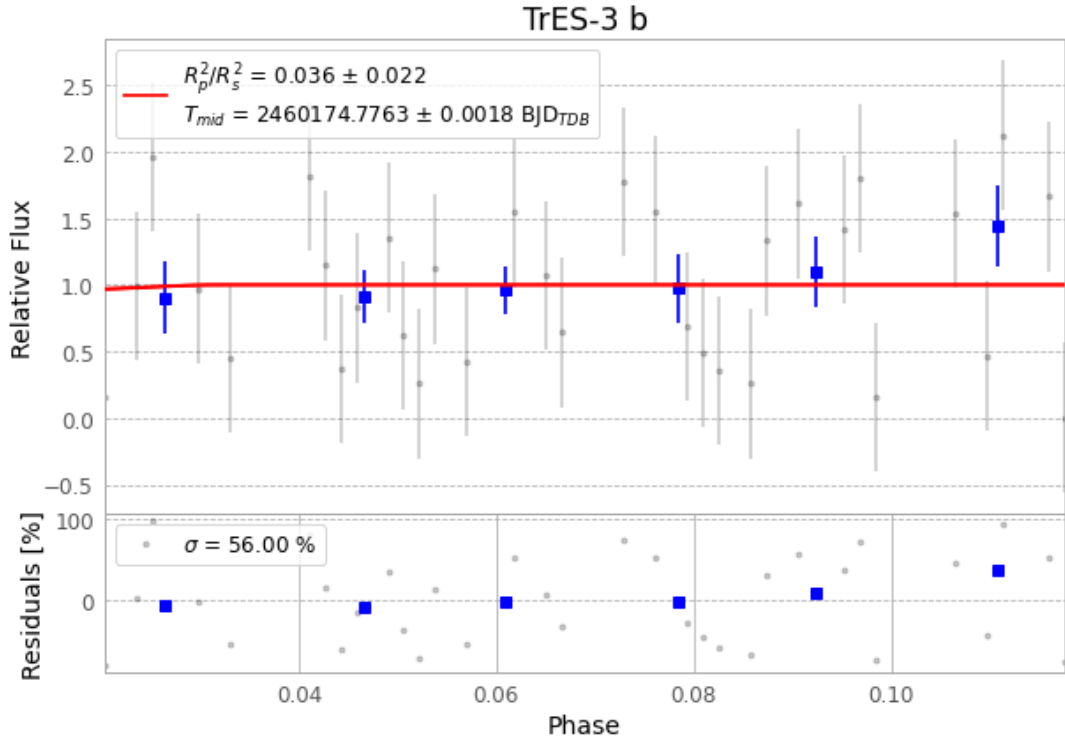


Figure 1 — FinalLightCurve_TrES-3 b_2023-08-18.png (SHA-256 44c6ff548a9c2254...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 efbb94d305e03718...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

63 FITS frames (41 MB), TRES-3230818070915.FITS → TRES-3230818101515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230818.log (c569cc6b77b1...), FinalLightCurve_TrES-3 b_2023-08-18.png (44c6ff548a9c...), FinalLightCurve_TrES-3 b_2023-08-18.csv (cd0a7f0e3e31...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 23:07Z.